ACDT 2026 · PROJECT 1 · GROUP 23 · PITCH SCRIPT (ABOUT 6½ MINUTES)

Fare Hike, Same Ride · pitch script

Six speakers (A–F) walking through the website from top to bottom. About 130 words a minute. Assign speakers A–F among the team; times are targets.

1 · Hook 0:00–0:30 · Speaker A

Screen: hero section, the 33% ticket.

> Seoul raised its subway fare by eighteen percent in 2023. If you commute, would you switch to a 따릉이 bike? Most people say no, because 따릉이 is for leisure, not for getting to work. We put that belief to the test with 680,000 station-days of rental data. Here is the first surprise: one in three weekday rides starts in just four commuting hours.

2 · The investigation 0:30–2:30 · Speakers B and C

Screen: chapter 01 (two claims, hypothesis), then chapter 02 (design table).

> B · The conventional wisdom makes two claims. A premise: bikes are a leisure mode. And a conclusion: so fares do not change how they are used. We tested both, because a conclusion can be right for the wrong reason. Before seeing any data we fixed our rule: a two-sided test, and a change counts only if it is at least three percentage points.

> B · The key idea was to measure what people ride for, not how much they ride. Weather can change daily rides six-fold. So our outcome is the share of weekday rides that start in commuting hours. We compared the same calendar dates one year apart, around two fare rises: plus eighteen percent in 2023 and plus five percent in 2025.

> C · Getting the data was the hard part. Twelve hourly files, up to 450 megabytes each, were too big for our Google Drive, so we condensed them one at a time in Colab. Excel cut the December 2023 file at its row limit, so we parsed it without Excel. And the subway-station coordinates had no download, so we built our zones from station names: 153 subway-exit stations and 147 park stations. Our November 2022 numbers matched our pilot exactly.

3 · The evidence 2:30–4:30 · Speakers C and D

Screen: chapter 03 model toggle and day calculator; chapter 04 scrollytelling, test cards, test calculator; chapter 05 causal chart.

> C · First, weather. Our professor asked why we used simple regression. So we built a multiple regression step by step. Temperature alone explains 46% of daily rides; adding rain, rest days and the curve explains 86%. And leaving rain out had hidden part of the heat effect, because hot days in Seoul are also the rainy days. Watch the calculator: same temperature, add ten millimetres of rain, and the forecast drops by tens of thousands of rides.

> D · Now the fare. Each dot is one weekday. After the 2023 rise, the commuting share moved +0.71 points, with a 95% interval from −0.58 to +2.00. After 2025, −0.79. Both intervals sit inside our three-point band, and an equivalence test confirms it. We ran ten tests, including multiple regression with weather and weekday controls, and they agree. You can change our choices in the calculator yourself.

> D · Is this causal? We compared the commute zone with park stations on the same days, compared each station with itself, and drew 2,000 random groups of stations as a placebo. Every estimate stays inside plus or minus three points. Our confidence: any fare effect is below about two points, and probably below one.

4 · The revelation 4:30–6:00 · Speaker E

Screen: chapter 06 myth-vs-reality tabs, chapter 07 maps (A, B, C), chapter 08.

> So the conclusion holds. But the premise does not. If 따릉이 were for leisure, people would ride more on weekends. They ride 25,726 fewer on rest days. Park stations are almost as commute-heavy as subway exits. What is true is subtler: commuting-hour rides rise and fall with the weather exactly like every other ride. 따릉이 is a fair-weather option, not a subway substitute. Our three system maps show the same picture from three angles, and the machine-learning map found something our regression missed: hours of rain matter more than millimetres.

> Our recommendation: do not expect fare rises to move commuters onto bikes. Plan bike supply around the weather forecast, and measure peak-hour dock availability and Climate Card use, because those hidden features could still be masking a small fare effect.

5 · The process 6:00–6:45 · Speaker F

Screen: chapter 09.

> Three lessons. Not significant is not the same as no effect, which is why we used an equivalence test. The unit decides the answer: counted per ride, a tiny change looks significant; counted per day, it does not. And controls move estimates, as our professor's question showed. Next, we want a no-fare-change placebo year and subway ridership by hour. Thank you. Everything, including the data and code, is downloadable from our site.